Indices, Surds and Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

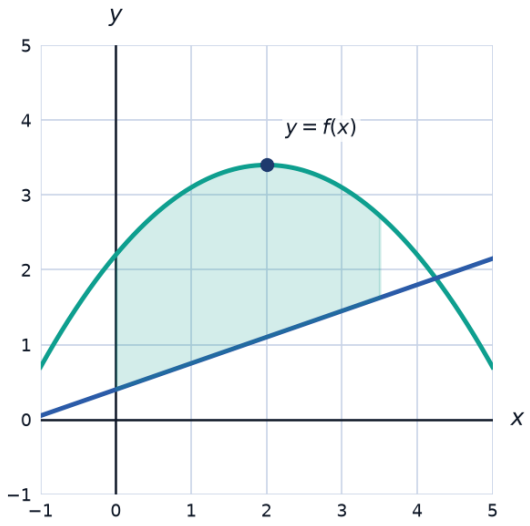
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

At 14:00, a coastguard radar station detects two vessels, A and B. Relative to the radar station, vessel A has position vector $(-4\mathbf{i} + 6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Vessel B has position vector $(20\mathbf{i} - 4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 10\mathbf{j})$ km h^{-1}, where u is a constant. Given that vessel B intercepts vessel A, find the value of u.

2 [Foundational] • Target: Fractional Indices and Exponential Equations

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.



3 [Foundational] • Target: Surds and Rationalising Denominators

Given that ${}^{n}\text{P}_{3} = 8\,{}^{n}\text{C}_{2}$, find the value of the integer n, where $n \geq 3$.

4 [Foundational] • Target: Remainder and Factor Theorems

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.

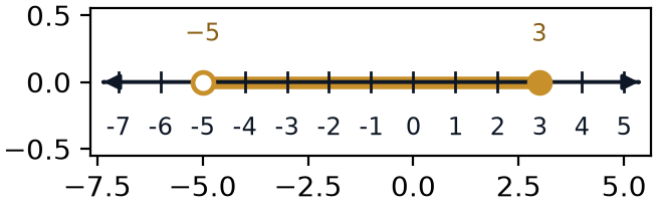
5 [Foundational] • Target: The Binomial Theorem

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm?

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6 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Find all solutions of $4\cos^2 x - 3 = 0$ in the interval $0 \le x \le 2\pi$.



7 [Foundational] • Target: Fractional Indices and Exponential Equations

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Surds and Rationalising Denominators

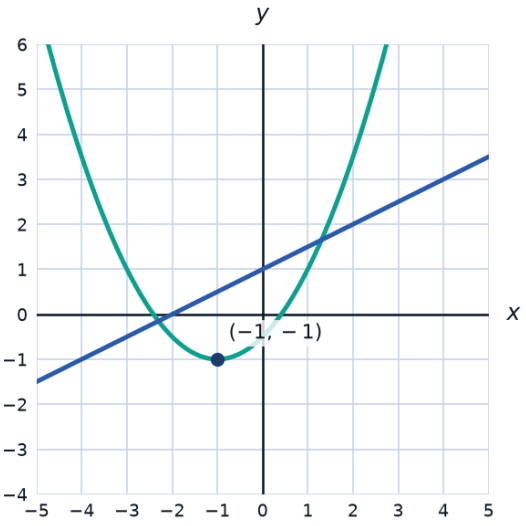
Evaluate the definite integral: $\int_0^1 (6x + 1)^3 \, dx$.

9 [Intermediate] • Target: Remainder and Factor Theorems

A sector of a circle of radius 12 cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

10 [Intermediate] • Target: The Binomial Theorem

In the expansion of $\left(1 + \frac{x}{4}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?



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11 [Intermediate] • Target: Indices, Surds & Polynomials Practice Exercises

The polynomial $P(x)$ is defined by $P(x) = (2 + ax)^n + bx^4 + cx - 4$ where n is an integer greater than 2, and a, b, c are constants with $a > 0$. In the binomial expansion of $(2 + ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x + 2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x + 1)$ is 7, find the remainder when $P(x)$ is divided by $(x - 1)$.

12 [Intermediate] • Target: Fractional Indices and Exponential Equations

In the expansion of $(1 + ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a > 0$. A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x - a)^2$ as a factor and leaves a remainder of n when divided by $(x - 1)$. Find the remainder when $P(x)$ is divided by $(x + a)$.

13 [Intermediate] • Target: Surds and Rationalising Denominators

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants. Find the remainder when $P(x)$ is divided by $x - 3$.

14 [Intermediate] • Target: Remainder and Factor Theorems

In the expansion of $(1 + ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

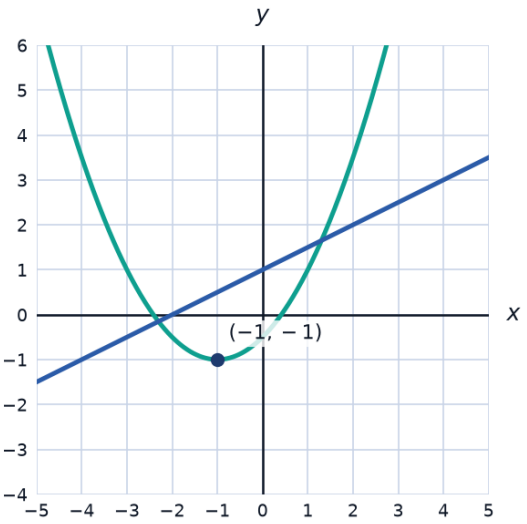
15 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

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16 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

When the polynomial $P(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a? Show your complete algebraic work.



17 [Challenging] • Target: Fractional Indices and Exponential Equations • Cross-Lesson: 6.01 Permutations and Combinations

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.

18 [Challenging] • Target: Surds and Rationalising Denominators • Cross-Lesson: 7.01 Differentiation and Rates of Change

In the expansion of $(1 + \frac{x}{4})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n? Show your complete algebraic work.

19 [Challenging] • Target: Remainder and Factor Theorems • Cross-Lesson: 8.01 Integration and Kinematics

In the expansion of $(1 + \frac{x}{5})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n? Show your complete algebraic work.

20 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In the expansion of $(1 + \frac{x}{6})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|2x + 3| = 7 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = -2 \) and \(x = -\frac{10}{2} \)

c. \(x = 2 \) only

d. \(x = 2 \) and \(x = -\frac{10}{2} \)

22 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|3x + 3| = 9 \).

a. \(x = 2 \) and \(x = -\frac{12}{3} \)

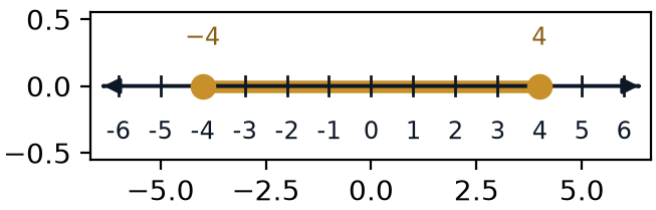
b. \(x = 3 \) and \(x = -3 \)

c. \(x = -2 \) and \(x = -\frac{12}{3} \)

d. \(x = 2 \) only

23 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|4x + 3| = 11 \).



a. \(x = 2 \) only

b. \(x = 3 \) and \(x = -3 \)

c. \(x = 2 \) and \(x = -\frac{14}{4} \)

d. \(x = -2 \) and \(x = -\frac{14}{4} \)

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24 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|5x + 3| = 13$.

a. $x = 2$ and $x = -\frac{16}{5}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{16}{5}$

d. $x = 3$ and $x = -3$

25 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|6x + 3| = 15$.

a. $x = 3$ and $x = -3$

b. $x = 2$ and $x = -\frac{18}{6}$

c. $x = 2$ only

d. $x = -2$ and $x = -\frac{18}{6}$

26 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. $x = 2$ and $x = -\frac{20}{7}$

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

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27 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|8x + 3| = 19 \).

a. \(x = 2 \) and \(x = -\frac{22}{8} \)

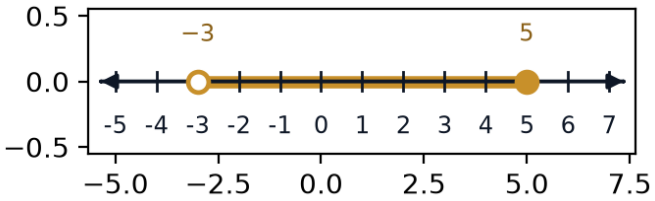
b. \(x = 2 \) only

c. \(x = -2 \) and \(x = -\frac{22}{8} \)

d. \(x = 3 \) and \(x = -3 \)

28 [Foundational] • Target: Indices, Surds & Polynomials Practice Exercises

Solve the modulus equation \(|9x + 3| = 21 \).



a. \(x = 2 \) only

b. \(x = 2 \) and \(x = -\frac{24}{9} \)

c. \(x = 3 \) and \(x = -3 \)

d. \(x = -2 \) and \(x = -\frac{24}{9} \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Fractional Indices and Exponential Equations

Evaluate the exact numerical value of $16^{-\frac{3}{4}} \times 27^{\frac{2}{3}}$.

a. $\frac{9}{8}$

b. $\frac{8}{9}$

c. $\frac{3}{4}$

d. $\frac{27}{16}$

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30 [Intermediate] • Target: Fractional Indices and Exponential Equations

Solve the exponential equation $2^{2x+1} - 9(2^x) + 4 = 0$ for all real x.

a. $x = -1$ and $x = 2$

b. $x = 1$ and $x = 4$

c. $x = 0$ and $x = 2$

d. $x = -2$ and $x = 1$

31 [Intermediate] • Target: Fractional Indices and Exponential Equations

Simplify the index expression $\frac{(4x^3y^{-2})^{\frac{1}{2}}}{2x^{-1}y^3}$.

a. $x^{\frac{5}{2}}y^{-4}$

b. $2x^{\frac{5}{2}}y^{-4}$

c. x^2y^{-4}

d. $x^{\frac{5}{2}}y^2$

32 [Intermediate] • Target: Surds and Rationalising Denominators

Rationalise the denominator and simplify $\frac{4+\sqrt{3}}{2-\sqrt{3}}$ into the form $a + b\sqrt{3}$.

a. $11 + 6\sqrt{3}$

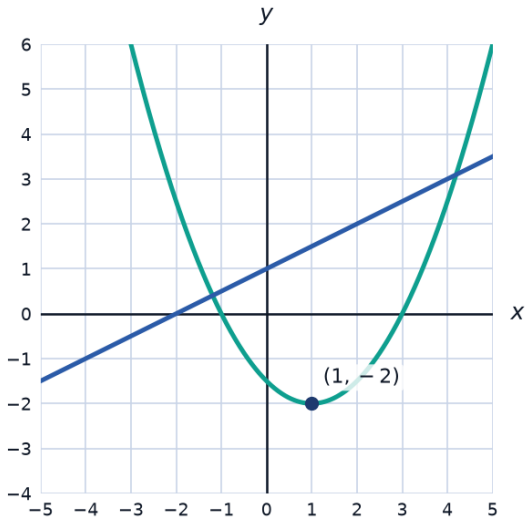
b. $5 + 6\sqrt{3}$

c. $11 - 6\sqrt{3}$

d. $7 + 4\sqrt{3}$

33 [Intermediate] • Target: Surds and Rationalising Denominators

Express $(3 - 2\sqrt{5})^2$ in the simplified form $p + q\sqrt{5}$.



a. $29 - 12\sqrt{5}$

b. $29 + 12\sqrt{5}$

c. $11 - 12\sqrt{5}$

d. $19 - 6\sqrt{5}$

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34 [Intermediate] • Target: Surds and Rationalising Denominators

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all valid real solutions.

a. $x = 1$ only

b. $x = 1$ and $x = -3$

c. $x = 3$ only

d. No real solutions

35 [Intermediate] • Target: Remainder and Factor Theorems

When $P(x) = 2x^3 + ax^2 - 5x + 6$ is divided by $(x - 2)$, the remainder is 16. Find the value of a.

a. $a = 1$

b. $a = -2$

c. $a = 3$

d. $a = 4$

36 [Intermediate] • Target: Remainder and Factor Theorems

Given that $(2x - 1)$ is a factor of $f(x) = 2x^3 + 3x^2 - 8x + c$, find the constant c.

a. $c = 3$

b. $c = -3$

c. $c = 6$

d. $c = -2$

37 [Intermediate] • Target: Remainder and Factor Theorems

Factorise completely the cubic polynomial $g(x) = x^3 - 4x^2 - 7x + 10$.

a. $(x - 1)(x + 2)(x - 5)$

b. $(x + 1)(x - 2)(x + 5)$

c. $(x - 1)(x - 2)(x - 5)$

d. $(x + 1)(x + 2)(x - 5)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the coefficient of x^3 in the binomial expansion of $(1 + 2x)^6$.

a. 160

b. 80

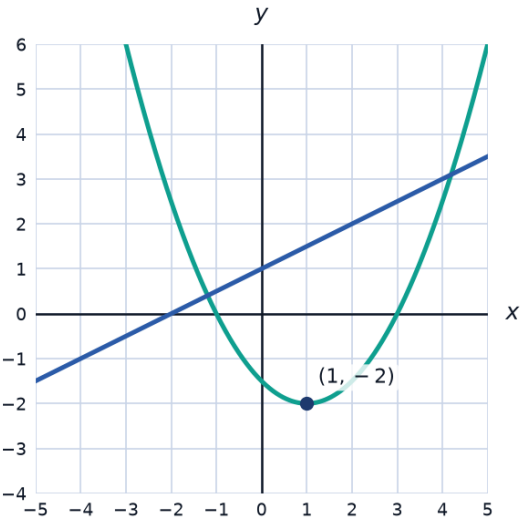
c. 240

d. 120

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39 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 8.01 Integration and Kinematics

Find the term independent of x (the constant term) in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.



a. 240

b. 160

c. 60

d. 192

40 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In the expansion of $(1 + ax)^5(1 - x)$, the coefficient of x is 9. Find the value of a.

a. $a = 2$

b. $a = 1$

c. $a = 3$

d. $a = -2$

41 [Challenging] • Target: Indices, Surds & Polynomials Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In the expansion of $\left(1 + \frac{x}{7}\right)^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

a. 29

b. 31

c. 28

d. 15

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42 [Challenging] • Target: Fractional Indices and Exponential Equations • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

In the expansion of $(1 + \frac{x}{8})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

a. 17

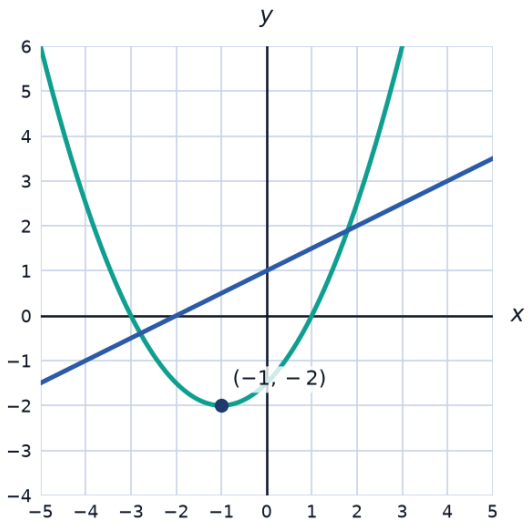
b. 33

c. 35

d. 32

43 [Challenging] • Target: Surds and Rationalising Denominators • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

In the expansion of $(1 + \frac{x}{3})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?



a. 12

b. 7

c. 15

d. 13

44 [Challenging] • Target: Remainder and Factor Theorems • Cross-Lesson: 6.01 Permutations and Combinations

In the expansion of $(1 + \frac{x}{2})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

a. 9

b. 11

c. 8

d. 5

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45 [Challenging] • Target: The Binomial Theorem • Cross-Lesson: 7.01 Differentiation and Rates of Change

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 6)$. What is the value of the constant a?

a. -14.33

b. 11.33

c. -11.33

d. -9.33

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